(vii) Deleterious Materials – Soil shall be virtually free from subsoil, wood including woody plant parts, toxic materials, stones over 30 mm, foreign objects, and propagules of plant species designated as noxious under the Weed Control Act & Weed Control Regulation, and other invasive or undesirable plant species, as determined by the Ministry Representative,

(viii) Drainage – in accordance with Table 751-A. Drainage of growing medium can be measured only after growing medium is in place. Mixing and handling of growing medium shall be done in such a manner that the minimum saturated hydraulic conductivity indicated is achieved.

751.17 Soil Amendments – The following soil amendments shall be added to the topsoil as required.

751.17.01 Peat Moss – Peat moss shall be Horticultural grade, partially decomposed fibrous or cellular stems and leaves of sphagnum mosses with a texture varying from porous to spongy fibrous, fairly elastic and substantially homogeneous with pH value not less than 3.5 and not greater than 6.5, medium to coarse shredded, suitable for horticultural purposes.

751.17.02 Sand – Sand shall be clean river pump sand or alternative source approved by the Ministry Representative, free of impurities, chemical or organic matter, and meeting the gradation specified in Table 751-B.

Table 751-B: Sand Particle Size Gradation

Sieve Size (mm)	Percent Passing (%)
4.75	95 – 100
0.600	0 – 40
0.075	0 – 5

751.17.03 Manure and Compost – Manure shall be well-rotted farm animal manure or mushroom manure, rotted to the extent that the material is crumbly. Manure shall be free from propagules of plant species designated as noxious under the BC Weed Control Act & Regulation, and other invasive or undesirable plant species, rocks, sticks, rubble, and shall contain not more than 40% composted sawdust, straw or shavings.

Commercial compost shall be free from propagules of plant species designated as noxious under the BC Weed Control Act & Regulation, and other invasive or undesirable plant species, as determined by the Ministry Representative, coliform, pathogens and chemical or toxic contaminants. Physical contaminants such as rocks, plastic, metal or glass shall be less than 0.5%. Compost shall not be derived from or contain processed municipal sewage sludge (i.e. biosolids), unless such product is authorized for use by Provincial Environmental Agencies, and meets all local regulations and approvals.

751.17.04 Wood Residuals – Raw sawdust and wood waste are not acceptable components of topsoil. Wood residuals used as a component of topsoil, compost, farm animal manure or mushroom manure are acceptable provided they are rotted and the total Carbon to total Nitrogen ratio for the topsoil is a maximum of 40:1.

751.18 Fertilizers – Fertilizers shall be standard commercial brands, meeting the requirements of the Canada Fertilizers Act and Fertilizers Regulations.

All fertilizers shall be in granular, pelleted or prill form, and shall be dry, free-flowing and free from lumps.

Fertilizers shall be packed in standard waterproof containers, clearly marked with the name of the manufacturer, weight and guaranteed analysis.

All fertilizer shall be stored in a weatherproof storage place and in such a manner that it will stay dry and its effectiveness will not be impaired.

The types, formulations, and rates of application for fertilizers and liming agents to topsoil supplied by the Contractor shall be as recommended by a laboratory soil specialist on the basis of tests of the topsoil, and as approved by the Ministry Representative.

Substitutions or variations in fertilizers and methods shall be made only upon pre-approval by the Ministry Representative.

751.19 Fill Material – Fill Material shall not be toxic to plant and animal life in part or in concentration and may not contain invasive knotweed plant material (*Polygonum* or *Fallopia* spp.).

CONSTRUCTION

751.31 Area Preparation

751.31.01 Stripping of Topsoil – Existing top soil material, shall be stripped and removed to stockpile(s) within the project area at least 10 m from water unless otherwise approved by the Ministry Representative, kept properly drained, and maintained in a neat and presentable condition free of spoil, propagules of invasive plants and other weedy species and subsoil material for subsequent spreading on prepared rough graded areas.

Sites chosen for storage of topsoil shall be free of noxious weeds and invasive plants. If invasive plants or noxious weeds are present at a storage site, propagules shall be removed prior to stockpiling topsoil at the site.

If topsoil is planned to be or actually is stockpiled for more than one month, those stockpiles must be covered or seeded to prevent erosion and invasive plant or noxious weed establishment.

Stockpiled topsoil shall be inspected for the presence of invasive plants or noxious weeds prior to spreading onto prepared rough graded areas, and any topsoil containing noxious weeds shall either be treated prior to use to remove all propagules or disposed of appropriately at the contractor's expense, unless the Ministry Representative authorizes its use as fill elsewhere on the project. Topsoil containing knotweed plant material (*Polygonum* or *Fallopia* spp.) shall not be used as fill.

The storage of topsoil shall not interfere with the effective utilization of a granular source or borrow pits. See also SS 165.06.02 Placement of Stripped Material.

751.31.02 Preparation of Landscape Area Subgrade – This Section applies only to grading of landscaped areas outside the roadway prism.

Rough grading shall be carried out by necessary cutting and filling work to produce the lines and grades shown on the Drawings and as directed by the Ministry Representative, allowing for the stipulated new topsoil thickness.

Surplus excavated material shall be removed from the Site and disposed of at the Contractor's expense unless the Ministry Representative authorizes its use as fill elsewhere on the project.

The landscape area subgrade shall be prepared to a consistent 80 – 85% Standard Proctor maximum dry density in accordance with ASTM D698.

Soft and unstable areas below the landscape area subgrade that cannot be compacted to this standard shall be excavated and filled with suitable fill material, except in locations where special environmental conditions have been identified. In such cases, appropriate alternative solutions shall be approved by the Ministry Representative and Environmental Agencies as required and carried out.

Debris, roots, branches, stones, building material, contaminated subsoil, visible invasive plants or weeds and anything else that may interfere with the proper growth and development of the planned finished landscaping shall be removed.

Fill materials shall be placed so as to achieve stability. This may necessitate placing in lifts of 225 mm and compacting each layer to 80 – 85% Standard Proctor maximum dry density.

Grade transitions of landscape area subgrade should be smooth and even and shall be such that ponding cannot occur on the landscape area subgrade surface.

Existing landforms shall be warped and blended into the landscaped areas with a minimum of visual disharmony.

751.31.03 Fine Grading – Areas requiring topsoil shall be fine graded by raking out spoil material and debris such as rocks, asphalt and concrete over 50 mm in diameter.

Naturalized areas not requiring topsoil shall be similarly cleaned, raked and manicured.

751.31.04 Scarifying – All landscape area subgrade shall be scarified to a minimum depth of 150 mm perpendicular to the slope immediately before placing topsoil.

751.31.05 Cleanup – All unsuitable material and inorganic debris shall be removed from the project area by the Contractor unless the Ministry Representative authorizes its use in fill areas on the project.

All surplus or unsuitable organic waste and debris shall be removed from the Site and disposed in a suitable location unless its complete burning is approved by the Ministry Representative in compliance with the B.C. Open Burning Smoke Control Regulation.

751.32 Placing Topsoil – The landscape area subgrade shall be inspected and approved by the Ministry Representative, before topsoil is placed.

751.32.01 General – The topsoil shall be:

(a) placed over the prepared landscape area subgrade and shall be allowed to settle or be compacted by light rolling such that it is firm against deep footprints and shall not be compacted more than necessary to meet this requirement;

(b) moist (25% to 75% of field capacity) but not wet when placed, and shall not be handled if frozen or wet such that its structure will be altered;

(c) manually spread around trees, shrubs and obstacles;

(d) evenly spread to a depth which after settlement or light compaction will be that shown in the Drawings or as directed by the Ministry Representative.

751.32.02 Cleanliness – During hauling and spreading, the paved roadway and other finished surfaces including subgrade under future base courses shall be kept clean and free of all topsoil.

751.32.03 Manufactured Soil – Placement of manufactured soil shall follow the manufacturer's specifications.

751.33 Applying Fertilizers – Fertilizers and soil amendments shall be added to bring topsoil fertility within the ranges set out in this Section or as specified in the Special Provisions.

Manufactured topsoil and processed imported topsoil will typically have fertilizers and amendments incorporated at the time of mixing and screening, while other topsoil will receive in situ amendment.

Fertilizers normally applied at the time of seeding and planting are specified under the appropriate sections. These fertilizers are generally supplemental to the base fertility requirements outlined for topsoil and are applied after topsoil is in place.

Surface applied fertilizers shall be evenly spread over the topsoil with a suitable mechanical spreader and fully incorporated to a minimum depth of 50 mm.

Lime used for top dressing shall be thoroughly cultivated into the top 100 mm of topsoil.

751.34 Finish Grading – After placing the topsoil to the finish elevations and contours required, the grade shall be finished to a high standard, to the grades shown on the Drawings with a smooth and even surface. Rough spots and low areas shall be eliminated to ensure positive surface drainage, and the surface shall be left smooth, uniform, free of debris and firm enough to resist deep footprinting.

Topsoil placed in traffic islands and medians shall be crowned for drainage, as shown on SS Drawing SP751-01.

MEASUREMENT

751.81 General – Measurement for the supply of topsoil by the Contractor will be by the CUBIC METRE in the truck at the point of delivery. Topsoil removed from stockpiles will be measured in place in the stockpile.

The placing of topsoil will be measured by the SQUARE METRE for the stipulated topsoil thickness, unless stated otherwise in the Special Provisions.

PAYMENT

751.91 General – Payment for TOPSOIL supplied by the Contractor will be at the Contract Unit Price bid per cubic metre. The Unit Price bid for topsoil supplied by the Contractor shall be accepted as full compensation for all handling, any necessary screening and testing of topsoil, and for delivery and off-loading on the area to be topsoiled.

Payment for TOPSOIL SPREADING will be at the Contract Unit Price bid per square metre. The Unit Price bid for topsoil spreading shall be full compensation for all labour and equipment required for the specified preparation, spreading, and finish grading of the topsoil, and for all incidental work not required to be separately paid for.

Separate prices may be included in the Contract to cover the area preparation work such as:

(a) clearing and grubbing,

(b) stripping existing topsoil and removal to stockpile,

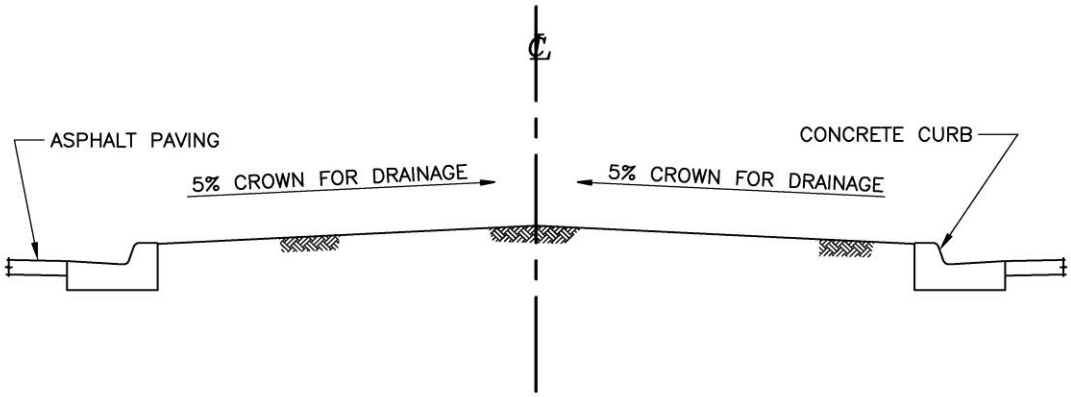
(c) rough grading and fill,

(d) naturalizing with fine grading,

(e) rotovating and soil conditioning, and

(f) trenching for irrigation and sub-soil drainage systems.

CROWNING DETAIL FOR ISLANDS & MEDIANS	SP751–01



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 754

PLANTING OF TREES, SHRUBS, AND GROUND COVERS

DESCRIPTION

754.01 Scope – This Section refers to those portions of the work that are unique to the supply and planting of trees, shrubs and ground covers, including seeded and sodded areas that are not designated for treatment under SS 757 "Revegetation Seeding". This Section must be referenced and interpreted simultaneously with all other Sections pertinent to the works described herein.

754.02 Related Work

- SS 751, Topsoil and Landscape Grading;
- SS 757, Revegetation Seeding.

754.03 References

- Canadian Nursery Stock Standard (Canadian Nursery Trades Association),
- BC Weed Control Act & Regulations,
- Integrated Pest Management Act & Regulation,
- Canada Seeds Act & Seeds Regulations,
- Fertilizers Act & Fertilizers Regulations,
- British Columbia Standard for Turfgrass Sod, and the
- Silviculture Manual (Ministry of Forests, Lands, Natural Resource Operations, and Rural Development).

754.04 Guarantee/Maintenance

754.04.01 Commencement – The Contractor shall guarantee and maintain all materials and quality of work for a period of one full year, commencing when the following conditions have been met:

(a) the supply and installation of all plant materials have been completed as per SS 754.43,

(b) all seeding/sodding has been completed (but is not necessarily yet established), and

(c) installation and hydrostatic testing of the irrigation system have been completed as per SS 766.42, and the system is fully operational.

754.04.02 Scope – The guarantee includes replacing all plants as determined by the Ministry Representative which are found dead or failing during the entire maintenance and guarantee period. Replacements shall be made immediately, unless otherwise directed by the Ministry Representative, and conditions of the guarantee shall apply to all replacement seeding for one full growing season.

Approval of plant material at the source does not preclude rejection of non-conforming stock on the Site prior to, or after planting.

The guarantee shall not apply to plants and planted areas damaged after Acceptance by causes beyond the Contractor's control, such as vandalism, "acts of God", "excessive wear and tear", or abuse.

MATERIALS

754.11 Source Quality Control – The Contractor shall notify the Ministry Representative of the sources of plant materials to be supplied at least two weeks prior to commencing the landscape work. All non-seed plant materials shall be made available for inspection at the nursery by the Ministry Representative upon at least three (3) days notice. Field grown material is not to be dug prior to inspection. All materials shall be free of plant species designated as noxious under the BC Weed Control Act & Regulation, and other invasive or undesirable plant species as determined by the Ministry Representative.

Approval of plant material at the source does not preclude rejection of non-conforming stock on the Site prior to, or after planting.

Imported plant material shall be accompanied by all necessary permits and import licences and shall conform to federal and provincial regulations.

It shall be the Contractor's responsibility to ensure that all regulations pertaining to the import of plant materials or their movement to or from a particular region of the province are adhered to, and all inspection certificates required by the Ministry of Agriculture are completed to the satisfaction of that Ministry.

754.12 Transporting Plants to Site

754.12.01 Dormant Period

(a) **Deciduous Bare Root Stock** (only in dormant period) – Adequate protection shall be given in order to preserve moisture around the root system. For short transit period, four hours or less, maximum temperature in the truck shall not be above 20°C. In all cases, at all times, roots should be protected from frost, wind and sun (e.g., a closed van with wet straw or other suitable packing material protecting the roots). The temperature shall be maintained as uniformly as possible by mechanical means, or in any event to prevent frost damage to roots. The appropriate temperature range shall be between 1°C and 10°C.

(b) Evergreens – It is recommended that root balls not be subjected to freezing temperatures below -5°C for a period longer than four hours. Plants shall be protected to prevent desiccation by wind and sun.

754.12.02 Non-Dormant period – Deciduous and Evergreen plant material shall be transported in a closed van or well-covered truck with a tarp or similar material in order to protect the leaves or needles from windburn. When in transit, with protection of a tarp cover only, it is recommended that foliage be sprayed with an antidesiccant. For the above material in transit for more than three days, it shall be unloaded and then stored away from direct sun for 24 hours to avoid leaf burning.

754.13 Unloading and Handling Procedures – Plants shall be carefully handled to minimize disturbance to root systems and damage to stems and branches. Plants shall not be dropped to the ground when unloading. Plant material that is mishandled and showing evidence of damage to root balls, or undue damage and breakage to top growth will be rejected.

Plants are to be kept in a moist condition at all times. All plants shall be well protected against physical damage and desiccation until they are planted on the Site.

754.13.01 B.R. (Bare Roots) Stock – Roots shall be covered and protected immediately from frost, sun and wind.

754.13.02 Stock in Pots/Containers – Shall be handled as much as possible by pots only in order to reduce breakage of branches/leaves.

754.13.03 Balled & Burlapped (B & B) Material – Plants shall be handled by holding the root ball and supporting the stem to minimize disturbance to the root ball and damage to stems and branches.

754.13.04 Material in Wire Basket – Specimen trees shall be lifted and supported by the wire basket and not by the trunk.

All plants shall be promptly unloaded, and their condition checked immediately upon arrival. Watering shall be provided as required and necessary pruning of minor breakage on branches performed.

754.13.05 Protection Against Stem and Branch Damage – During loading, transportation, off-loading and planting, all trees shall be protected against damage to stems and branches. This applies particularly to larger wire-basketed trees.

Bark shall be protected against chafing and cuts by providing a wrapping of cardboard, sackcloth or other material as appropriate and when required.

754.13.06 Sod – Sod shall be protected during transportation to prevent drying out and shall arrive at the site in a fresh and healthy condition.

754.14 Storage

754.14.01 Storage During Growing Season – All plants in containers, balled and burlapped, or in wire basket, if not planted within three days, shall be stored in an upright position in an area providing even light and offering protection from wind and sun scald. Enough space shall be provided between plants so that light reaches all around to the bottom of the plant in order to avoid leaf burning when planted out.

(a) Sod – Sod shall be installed as soon as possible after delivery. If there is a delay of more than 24 hours, the sod shall be properly stored and kept moist and cool until it is placed. Sod shall not be stacked more than three levels high while being stored.

(b) Balled and Burlapped Material – Special attention shall be given to the root ball, and unless weather is rainy or cool, root balls shall be protected by covering with material suitable to protect them from drying out (e.g. sawdust, peat moss, topsoil). Plants intended to be planted in the open shall not be kept stored in a building or any area of low light intensity for a period exceeding seven days. All plants shall be kept well watered and protected from heat and frost.

(c) Containerized Plants – In extreme weather, freezing or high dry heat, the containers shall be buried in a protective medium. Plants intended to be planted in the open shall not be kept stored in a building, truck or any area of low light intensity for a period exceeding seven days during the growing season.

754.14.02 Storage During Dormant Period – Plants shall be cared for according to each plant's requirement for winter protection, and according to geographical location.

754.15 Plant Material

754.15.01 Species – Selection of species shall be as specified. Every effort shall be made by the Contractor to obtain the plant material specified. Where evidence is submitted that a specified plant cannot be obtained, substitutions in kind, size and grade shall be made upon written approval by the Ministry Representative.

754.15.02 Origin and Requirements – All plant material shall be nursery grown stock or approved collected native plants unless specified otherwise. Plant material will be inspected by the Ministry Representative upon delivery to the Site.

All nursery grown plants shall, as a minimum, comply with the Canadian Nursery Landscape Association Specification "Canadian Nursery Stock Standard" with respect to sizing, grading and quality.

Plants shall be true to name, type and form and shall be representative of their species and variety.

All plants shall be sturdy stock, with tree and shrub heights proportional to trunk caliper, overall plant width and size of root ball, as determined by the Ministry Representative. Plants that are weak and thin, and those showing effects of being grown too closely together or poorly maintained, will not be accepted.

Plants shall be vigorous and healthy with normal, well developed branches and good fibrous root systems and be free from decay, physical injury, disease and insect damage and infestation.

In particular, conifers shall have a healthy, single leader with well shaped whorls of vigorous, newly growing branches and shall exhibit natural growth habit characteristic of the species and variety. Trees sheared as Christmas trees are not acceptable.

The root balls of dug material and the soil of containerized plants shall be free from pernicious perennial weeds. All balled and burlapped plant stock shall be supplied in biodegradable root ball sacking.

The search area for plants shall include, but not necessarily be limited to, the provinces of British Columbia, Alberta and the states of Washington, Oregon, California (northern portion), Idaho and Montana. All plant material being sourced from milder areas shall be properly "hardened off" prior to shipping and planting.

754.15.03 Nursery Grown Stock – All plants specified "Container" shall be grown for the length of time necessary to permit the roots to fill and hold the soil within the container, as required by the Canadian Standards for Nursery Stock.

Similarly, all field dug material will show evidence of having been root pruned to encourage fibrous root system development and resulting in root balls that retain their integrity during handling.

Forestry seedlings shall be supplied and handled in accordance with the requirements of the Silvicultural Manual.

All plant material shall be appropriately identified and individually labelled with weatherproof tags. In the case of small containerized plants such as ground covers and bundled bare root seedlings, which are supplied in large numbers, labelling shall be limited to identification of group lots as permitted by the Ministry Representative.

754.15.04 Collected Plant Stock – The Contractor shall provide either permits or verification that permission was obtained for collecting native and/or introduced plant materials. Information shall be supplied on where, when and how collection was made.

All collected plant material shall have been grown and maintained in a nursery environment for a minimum of one growing season, unless, at the sole discretion of the Ministry Representative, certain species are approved for a lesser period of time. As for nursery grown stock, collected plants shall be held long enough prior to planting to allow roots to fill the container or the dug root ball and to retain the soil within.

Collected plant stock shall be appropriately labeled with weather proof tags for easy identification at the job site.

754.15.05 Seed – Seed quality and type shall conform to that specified in the Special Provisions.

All other requirements are per SS 757 Revegetation Seeding.

754.15.06 Sod – Sod shall be nursery grown, true to type and conform to the British Columbia Standard for Turfgrass Sod, and the general requirements of the Canadian Nursery Stock Standard.

Sod grade shall be as specified in the Special Provisions.

754.16 Water – Water shall be clean and potable and shall be supplied by the Contractor.

754.17 Fertilizer – Fertilizer shall comply with the provisions of the Canada Fertilizers Act and Fertilizer Regulations. Fertilizer shall be supplied to the specifications in the Special Provisions.

754.18 Bark Mulch – Bark mulch shall be sized 25 mm and minus, Douglas Fir or Hemlock bark chips and fines, or a combination of both types and of the quality used for decorative landscape mulching purposes. It should be free of chunks and sticks, dark brown in colour and free of all soil, stones, roots or other extraneous matter.

754.19 Backfill Topsoil – Backfill topsoil for planting operations shall conform to the requirements of SS 751 – Topsoil and Landscape Grading.

754.20 Other Materials – When required, various other materials such as soil amendments, erosion control products, hydraulic mulches, etc. shall be supplied to the specifications in the Special Provisions.

CONSTRUCTION

754.31 Scheduling – Work shall be scheduled to meet the milestone dates provided in the Special Provisions, and to ensure its execution meets the requirements of living plant material.

The work shall be co-ordinated with the schedule of other trades and be well integrated with other specific requirements such as Sediment and Drainage Management Drawings, which may be provided for any given project.

754.32 Pre-planting Operations – The plant material shall be approved by the Ministry Representative prior to installation. The Contractor shall ensure that all requirements of SS 754.11 through SS 754.15 have been met and that any minor damage to plant stock is taken care of through appropriate pruning or other measure. When directed by the Ministry Representative, the Contractor shall apply anti-

desiccant to large conifers and deciduous trees that are in leaf. Application will be in accordance with the manufacturer's instructions for the particular product.

All invasive plants on site shall be removed or controlled prior to planting, seeding or installation of sod.

754.33 Location of Planting – Locations, quantities and spacing of trees, shrubs, vines and groundcovers as shown on the Drawings shall be considered approximate and may be adjusted by the Ministry Representative to meet field and recommended micro-site conditions as specified in the Silviculture Manual. Tree numbers, spacings and locations will vary according to the Site conditions and amenities. The Contractor may adjust plantings to meet field conditions, with the approval of the Ministry Representative. Locations shall be staked as shown on the Drawings and verified on Site with the Ministry Representative prior to planting. If underground obstructions are uncovered, they shall be reported to Ministry Representative for resolution.

754.34 Area Preparation

754.34.01 Finish Grade Preparation – The Contractor shall verify that grades are correct. If discrepancies occur, the Ministry Representative shall be notified, and work shall be halted until otherwise instructed by the Ministry Representative.

754.34.02 Planting Beds and Grass Areas – Prepare planting beds and grass areas in accordance with SS 751 – Topsoil and Landscape Grading.

754.34.03 Planting Holes – Planting holes shall be dug in accordance with the specific requirements described below. The bottom of planting holes shall be scarified and loosened to a depth of 100 mm prior to placement of plants and backfill soil.

Subsoil, rocks, roots and extraneous material shall be removed from excavated material that will be used as planting backfill soil. Unsuitable or excess material shall be disposed of.

Holes dug by a mechanical tree spade shall have their sides scarified to loosen any compaction glazing caused by the blades. Planting holes shall be tested by filling with water. Inadequate drainage conditions permitting the retention of water in planting pits for more than 12 hours shall be reported to the Ministry Representative before proceeding with the work.

(a) **Free Draining Sub-Grade**: Where the subgrade and existing native surface soils are of good drainage and of a non-compacted nature, planting holes shall be excavated and prepared to allow the following depth of topsoil backfill underneath and around the root ball:

- **(i)** For plants up to and including 27 cm (#5) pot size – not less than 150 mm.
- **(ii)** For plants larger than 27 cm (#5) pot size – not less than 300 mm.

(b) **Poor Draining Sub Grade**: Where the subgrade and existing native surface soils are of poor texture and conditions are generally compacted, planting holes shall be excavated and prepared to allow the following depth of topsoil backfill underneath and around the root ball:

- **(i)** For plants up to and including 27 cm (#5) pot size – not less than 300 mm
- **(ii)** For plants up to 45 cm pot size – not less than 450 mm
- **(iii)** For tree root balls larger than 45 cm – not less than 600 mm

(c) **Severely Compacted Conditions –** If severely compacted conditions are encountered, and surface or ground water entering the excavations does not drain, the Contractor shall correct the problem by:

- **(i)** providing a means of sub-surface drainage;
- **(ii)** utilizing elevated planting techniques where some of the planting soil will be placed into a partial excavation and the remainder on the surface to meet the depth requirement for growing medium; or
- **(iii)** considering alternate planting sites. These alternatives shall receive prior approval by the Ministry Representative, as applicable to the Site.

754.35 Time of Planting – All planting operations shall be performed during the normal planting season for each type of material, and within the milestone dates provided in the Special Provisions, unless otherwise authorized in writing. During the specified timeframe, planting operations shall, as far as practicable, take advantage of soil and weather conditions favourable to the work. Late spring or early fall planting is recommended unless supplemental watering is available.

Planting shall occur when risk of frost or snowfall is minimal. Planting into frozen ground is not acceptable.

754.36 Planting Procedures – Trees and Shrubs

754.36.01 General Procedure

(a) Plants shall be installed so that after settlement they will be at the same planting depth they were at in the field or in containers. The soil mark on the stem is an indication of this, and it shall be flush with the finished level allowing for settling of the topsoil after planting and settlement. The entire root ball shall be covered with growing medium.

(b) Once the bottom of the planting hole is scarified and the initial lift of backfill topsoil is placed, the holes shall be

pre-watered and allowed to drain prior to installation of plant material.

(c) Plants shall be set plumb in the planting beds or in the centre of the pits except where the plant's character requires variation from this.

(d) The growing medium shall be placed in layers around the roots or ball, preferably by hand. Each layer shall be firmed to eliminate air void and ensure good soil contact with the roots. The process shall be carried out carefully to avoid injuring the roots or ball or disturbing the position of the plant.

(e) Trees requiring staking shall have support stakes placed carefully between the roots before backfilling. Specifications for tree supports are described in this SS 754.

(f) After the planting hole is filled with soil to ground level, the plant shall receive a thorough watering.

(i) A final backfill layer shall be applied to form a saucer-like berm around the circumference of the planting hole in order to catch and hold rainwater.

(ii) This rain basin shall be maintained until final acceptance of the work.

(g) Once planting and mulching is complete, the Site shall be cleaned of all excess soil, rock and debris.

754.36.02 Specific Planting Requirements:

(a) **Bare Root Stock** – The roots of bare root plant material shall be soaked in water prior to planting. During installation, the plant roots shall be evenly spread out over a cone of soil in the bottom of the hole, and the plant supported to the correct depth as backfilling takes place. The plant shall be gently shaken in a vertical motion to ensure that soil particles sift into the root system and establish close contact with the roots.

(b) **Container Stock** – Non-perishable, impervious containers such as plastic pots and tubs shall be removed from plants before planting. Once plants are removed from these containers, root systems shall not be disturbed with the exception of unraveling any roots starting to spiral around the root ball.

Bio-degradable containers such as peat or paper fibre pots shall not be removed before planting but shall be thoroughly soaked with water prior to placement in the holes. This will ensure that containers absorb subsequent watering and not repel it due to the presence of a dry barrier. The rim of such containers shall be not be exposed to the air, and when necessary, shall be removed after planting.

(c) **Balled and Burlapped Stock** – When backfill soil is placed to approximately two-thirds of the root ball height, the ties on the sacking shall be cut and the top portion of the burlap folded back carefully to avoid disturbing the integrity of the root ball. The sacking shall not be removed. The remainder of the hole will then be backfilled and firmed.

Where wire baskets are used to encase and support the root ball of supplied plant material, these shall not be removed. The top of the wire basket shall be cut away or completely folded back and buried without disturbing the integrity of the root ball.

(d) **Forestry Seedlings** – Planting of forestry seedlings shall be in accordance with the Silviculture Manual.

(e) **Tree Support –** All trees of a size requiring staking or guy wiring shall be supported in accordance with the details and instructions provided on SS Drawings SP754-04 through SP754-07. All hardware required shall be installed without damage to plants.

Trees that are dislodged during the contract period shall be uprighted and re-secured as required. Trees that are damaged shall be replaced by the Contractor at the Contractor's expense.

754.37 Seeding – Application of seed, fertilizer, and other materials shall be at the rates specified in the Special Provisions.

The requirements for construction shall be in accordance with SS 757 Revegetation Seeding.

754.38 Sodding

(a) The required fertilizer shall be applied at the rates specified in the Special Provisions and worked well into the topsoil prior to laying the sod.

(b) Sod shall be laid within 24 hours after delivery unless proper storage arrangements can be made.

(c) The sections of sod shall be laid close together with joints staggered. No open joints are to be visible, and no pieces are to overlap.

(d) Sod shall be laid smooth and flush with the adjoining grass areas, adjacent hard surfacing, and the tops of curbs and planting bed liners, unless otherwise shown on the Drawings. All necessary cutting shall be done using sharp implements.

(e) **Planting on Slopes:**

(i) On slopes of approximately 2.5H to 1V and steeper, the sod shall be laid lengthwise across the slope, and the material secured with wooden stakes driven flush with the sod at intervals not exceeding 0.5 metres. There shall be at least three stakes per individual sod piece.

(ii) On slopes of gradients between 2.5H:1V and 1.5H:1V, the bottom three rows of sod and every third subsequent row shall be secured with stakes.

(iii) For slopes steeper than 1.5H:1V, every sod course shall be staked.

(iv) Stakes shall be removed before pedestrian traffic is permitted on the sod.

(f) Sodded areas shall be rolled or suitably tamped to ensure a good bond with the topsoil, and then subsequently protected from heavy foot traffic or equipment travel.

(g) Unless otherwise indicated, sodded areas shall be evenly watered within 12 hours of installation, and with sufficient quantity to saturate the grass and the upper portion of the topsoil.

754.39 Watering – All trees, shrubs, groundcovers, vines, and designated grass areas, shall be watered immediately after planting, and regular watering shall continue as required for plant health until final acceptance of the work.

754.40 Pruning – Pruning shall be limited to the minimum necessary to remove dead or injured tissue and branches interfering with desirable growth habit and overall health of the plant.

Pruning shall be done in accordance with proper horticultural practice, using clean, sharp tools appropriate to the task and in a manner that preserves the natural character of the plant.

754.41 Mulching – When specified in the Drawings and Special Provisions, individual tree pits and planting beds shall be mulched.

Mulching of tree pits and planting beds shall be carried out after watering, to an even depth of 50 mm after settlement, unless otherwise specified on the Drawings and in the Special Provisions.

754.42 Clean-up – All plant containers and waste materials resulting from landscaping and planting operations shall be removed from the Site and appropriately disposed of.

754.43 Conditions for Acceptance – The Contractor shall ensure that the following conditions are met for all planted and grassed areas.

754.43.01 Conditions

(a) Topsoil quality, fertility levels, depths and surface conditions are as set out in the Drawings and Specifications;

(b) All plants are of the species and varieties specified and planted in the locations shown on Drawings;

(c) All plants are healthy and growing vigorously. Seeded grass areas are sufficiently established into the underlying growing medium, are free of thin and bare patches, free of invasive plants, and are relatively free of other weedy species: not more than 5% in lawn areas, and not more than 15% in rough grass areas;

(d) The water content in the topsoil, i.e. – when irrigation is provided, is to the satisfaction of the Ministry Representative;

(e) Trees are supported, as specified;

(f) Pruning is complete, in accordance with proper horticultural practice and to the satisfaction of the Ministry Representative;

(g) All planting beds and tree pits are free of invasive plants and other weedy species;

(h) Mulch is in place, as required;

(i) Unmulched areas are cultivated to leave a loose, friable, water-permeable surface, and;

(j) Maintenance procedures set out in SS 754.71 have been carried out.

MAINTENANCE

754.71 Maintenance – The following maintenance operations shall be performed from the time of landscape installation, until the earlier of the Actual Completion Date and expiry of the landscape maintenance period as defined in the contract documents.

754.71.01 Maintenance Operations

(a) When specified, water shall be applied in sufficient quantity and by appropriate method to maintain optimum soil moisture conditions for healthy plant establishment, without causing surface soil erosion.

(b) Invasive plant and weed control will be carried out, as required, to prevent competition with establishing planted material and to maintain the aesthetic appearance of landscaped areas. Invasive plants must be removed completely, unless otherwise stated in the Special Provisions or otherwise directed by the Ministry Representative.

(i) The presence of other weedy species in plantation beds, individual planting pits, and designated lawn areas, is limited to a maximum of 5% of the surface area at any given time, unless otherwise stated in the Special Provisions or otherwise directed by the Ministry Representative.

(ii) The use of herbicides for the control of invasive plants must be approved by the Ministry Representative prior to use and must be conducted under a confirmed Integrated Pest Management Plan, in accordance with the Integrated Pest Management Act and associated Regulations, and must be completed by a Certified Pesticide Applicator.

(iii) The use of herbicides for control of other non-invasive weedy species is not permitted.

(c) Mowing shall be carried out at regular intervals, as required, to maintain grass in the areas designated, and at the height(s) specified in the Special Provisions. Edges of areas designated "Lawn" shall be neatly trimmed. Excess clippings shall be removed immediately after mowing and trimming.

(d) Fertility levels in planted and grassed areas shall be maintained in accordance with the requirements of the plant material.

(e) Bark Mulch shall be maintained to the specified depth.

(f) For non-mulched areas, the soil surface shall be cultivated, as required, to keep it loose and friable.

(g) Insect and disease control shall be carried out as required. The Contractor shall secure all necessary Permits and possess a valid Pesticide Service License. When required, the Contractor shall provide a Management Plan to the Ministry and any approval agencies.

(h) Establishment pruning to encourage proper shape and health of plants by removing dead or broken and interfering branches and diseased or damaged tissue.

(i) Maintenance of tree stakes, guy wires and tree ties to prevent plant dislodgement and damage to trunk and branches.

(j) All plant material shall be alive and maintained in a healthy growing condition during the entire establishment period. Plant material which has died or is not healthy, and in the opinion of the Ministry Representative, does not perform its function, will be removed and replaced by the Contractor at the earliest opportunity, weather and season permitting. Grassed areas that show deterioration or bare spots shall be repaired immediately. Unless otherwise directed by the Ministry Representative, all repair and/or replacement shall be in accordance with the original specifications and requirements.

MEASUREMENT

754.81 Planting – The unit of measurement for plants will generally be PER PLANT, unless otherwise specified in the Special Provisions.

754.82 Seeding and Sodding – The unit of measurement for seeded and sodded areas will generally be by the SQUARE METRE, unless otherwise specified in the Special Provisions.

754.83 Mulch – The unit of measurement for supply and installation of bark mulch will generally be by the SQUARE METRE for the stipulated mulch thickness, unless otherwise specified in the Special Provisions.

PAYMENT

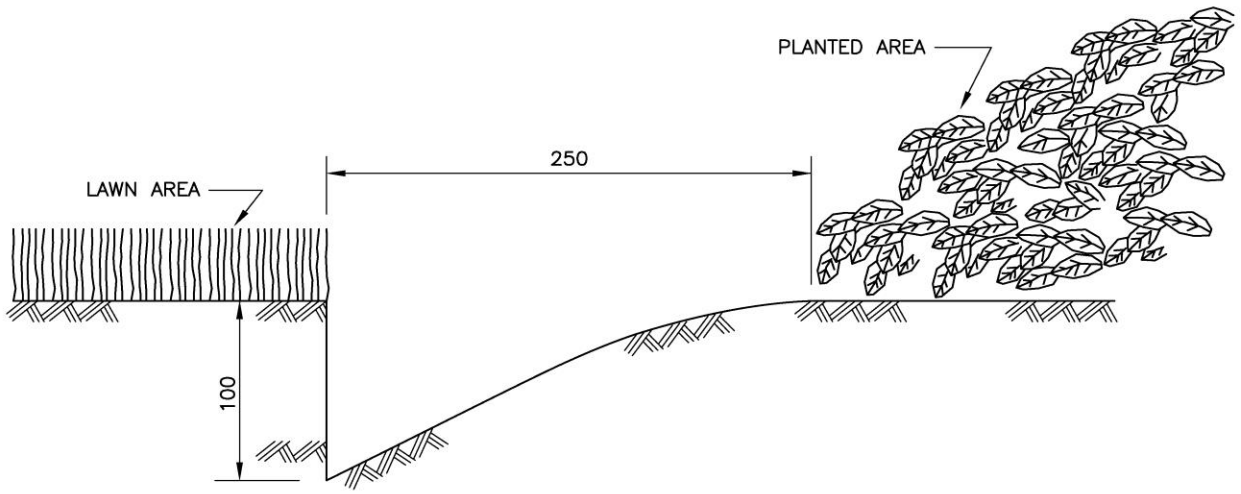
754.91 General

754.91.01 Planting – Payment for plants will be at the contract unit prices bid for PLANTING of the types, species and sizes called for, and shall constitute full compensation for supplying and delivering plants; for supplying and delivering topsoil for plant backfill, fertilizer and all incidental materials; for digging holes for plants; for planting, pruning, staking and guying, mulching, rain basin formation, clean up after planting and maintenance of plants; and for all labour, equipment and tools and incidentals necessary to complete the work prescribed in this Section.

754.91.02 Seeding and Sodding – Payment for SEEDING and SODDING will be at the contract unit price bid per square metre. The unit price shall be full compensation for work described and all work subsidiary and incidental thereto for which separate payment is not elsewhere provided.

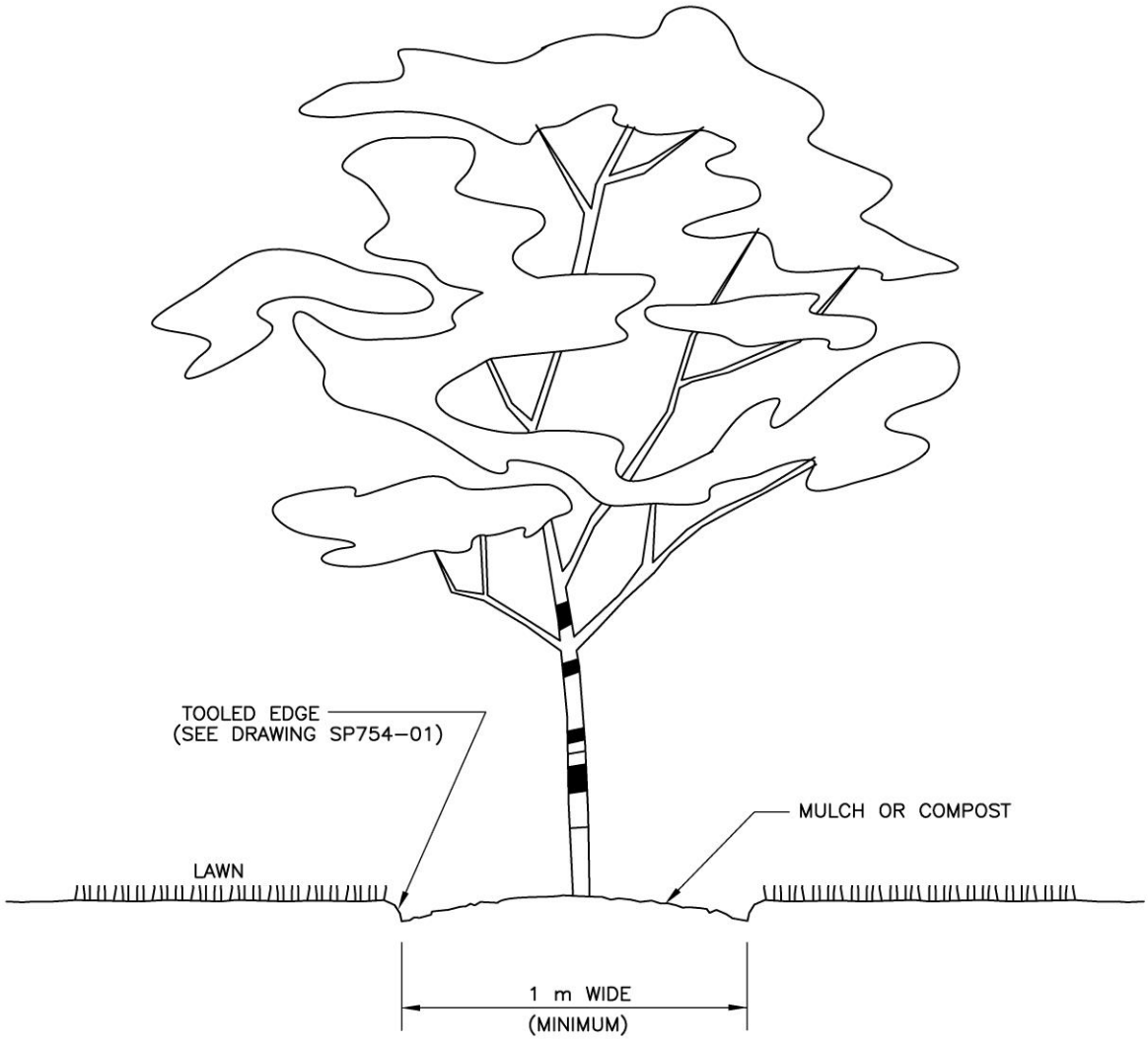
754.91.03 Mulching – Payment for MULCHING will be at the contract unit price bid per square metre. The unit price bid for mulching shall be full compensation for all labour and equipment required for supplying and spreading of the mulch and for all incidental work not required to be separately paid for.

PLANTATION EDGING DETAIL	SP754–01



NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

TREES WITHIN LAWN AREAS	SP754–02

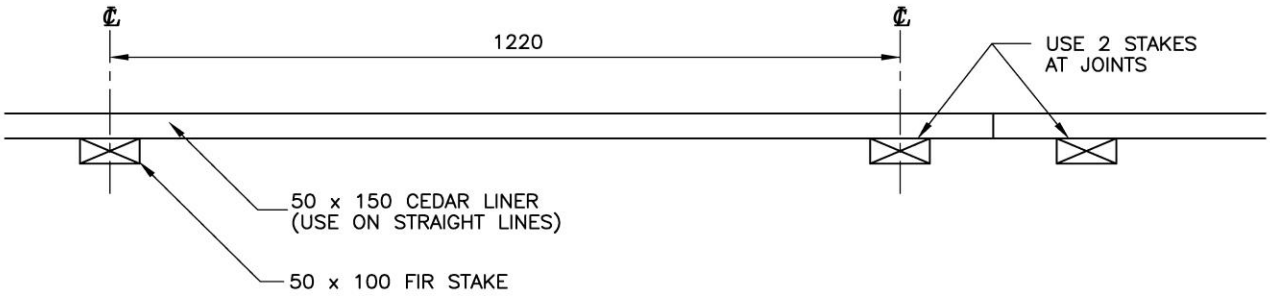


NOTE:
MAINTAIN VEGETATION FREE BASE

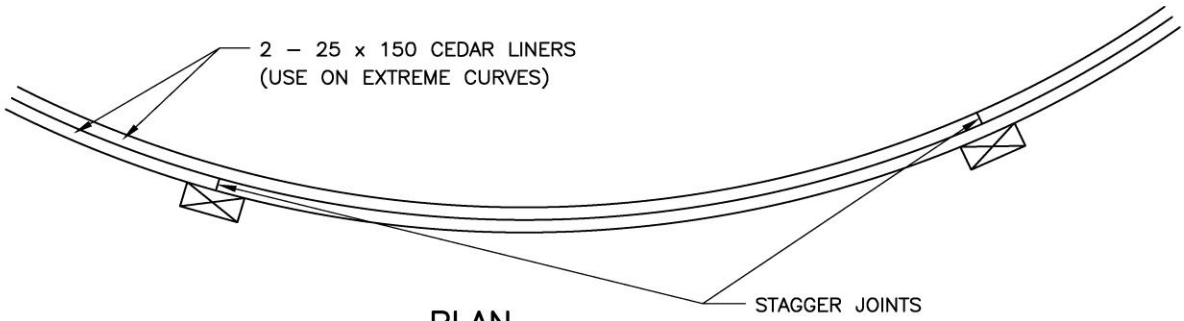
NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

CEDAR LINER DETAILS

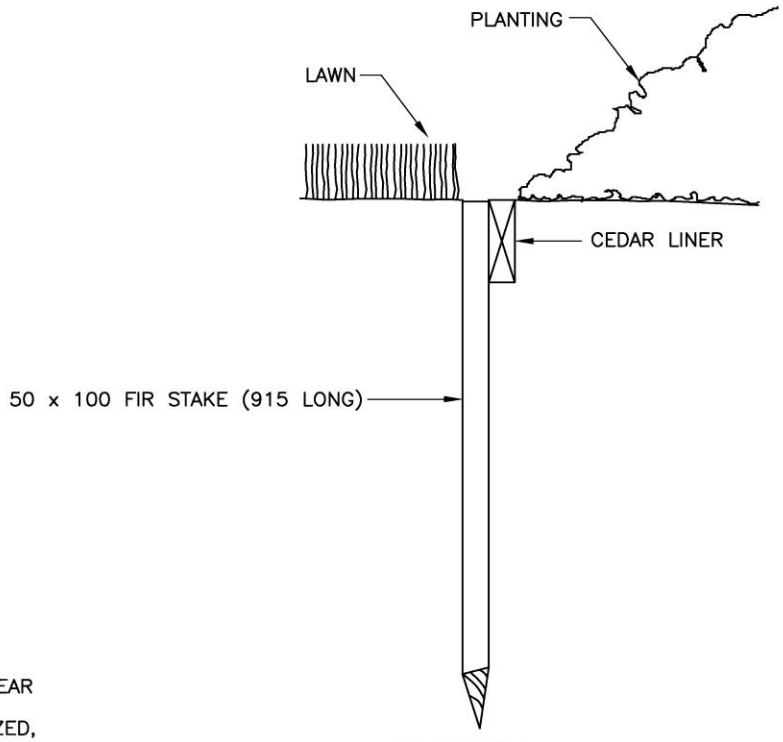
SP754–03



PLAN



PLAN



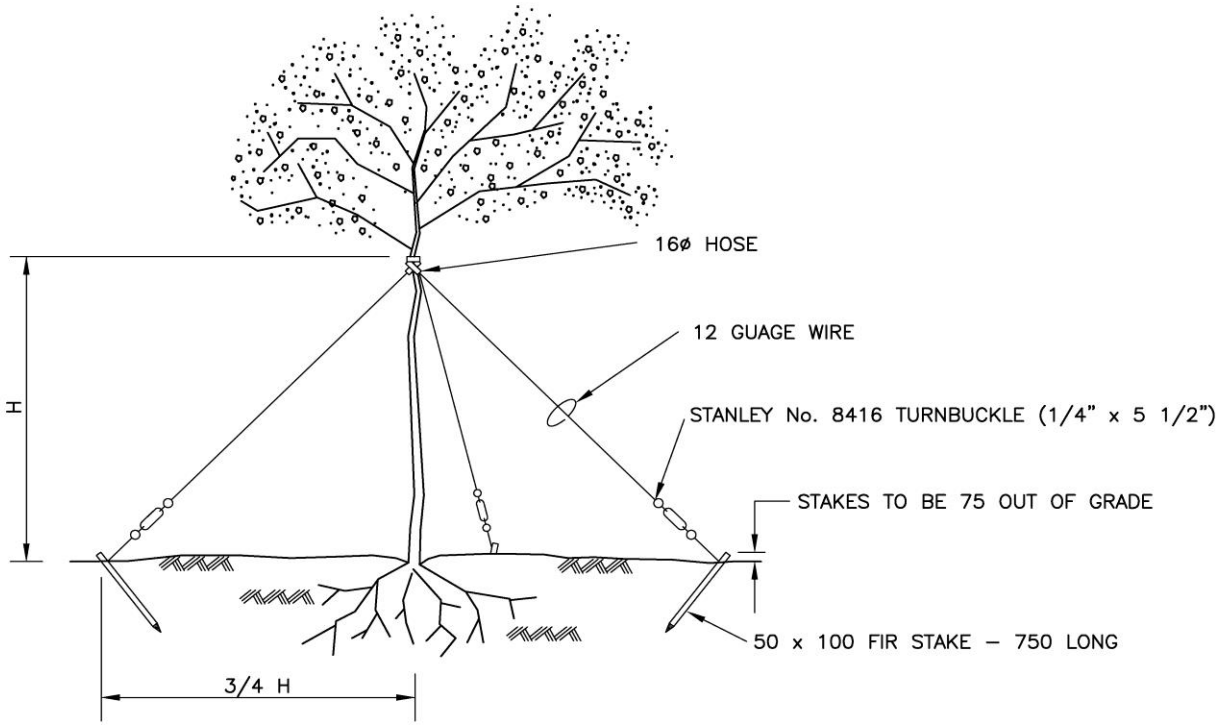
NOTES:
1. ALL WOOD TO RECEIVE 2 COATS CLEAR CREOSOTE PRIOR TO CONSTRUCTION
2. ALL NAILS TO BE HOT–DIP GALVANIZED, 75 mm LONG

SECTION

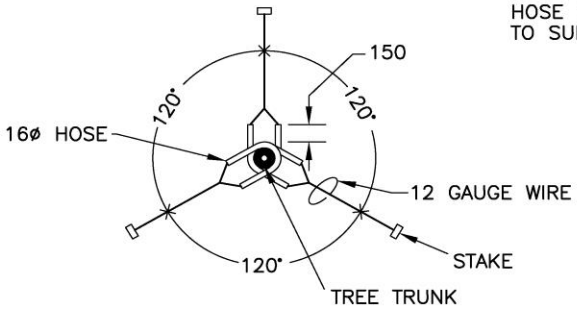
NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

TRIPOD BRACING DETAIL — SP754–04

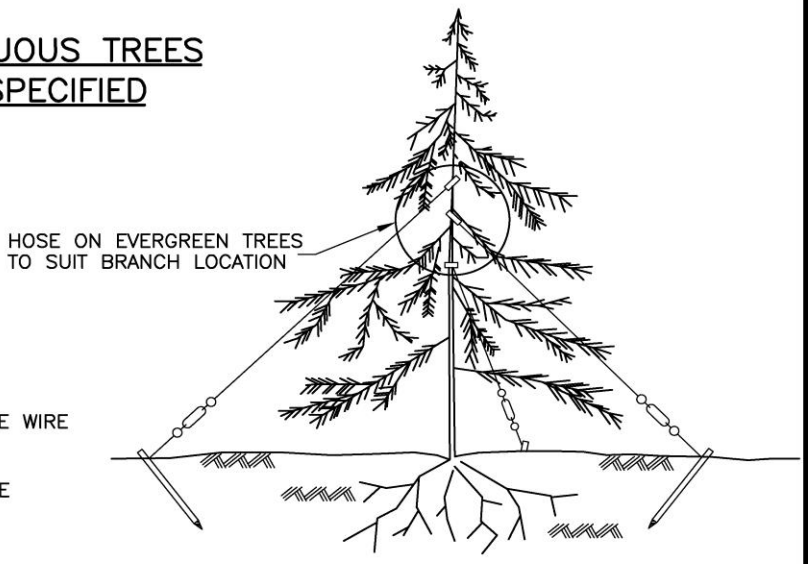


FOR DECIDUOUS TREES
WHEN SPECIFIED



PLAN VIEW

TURNBUCKLES NOT SHOWN FOR CLARITY

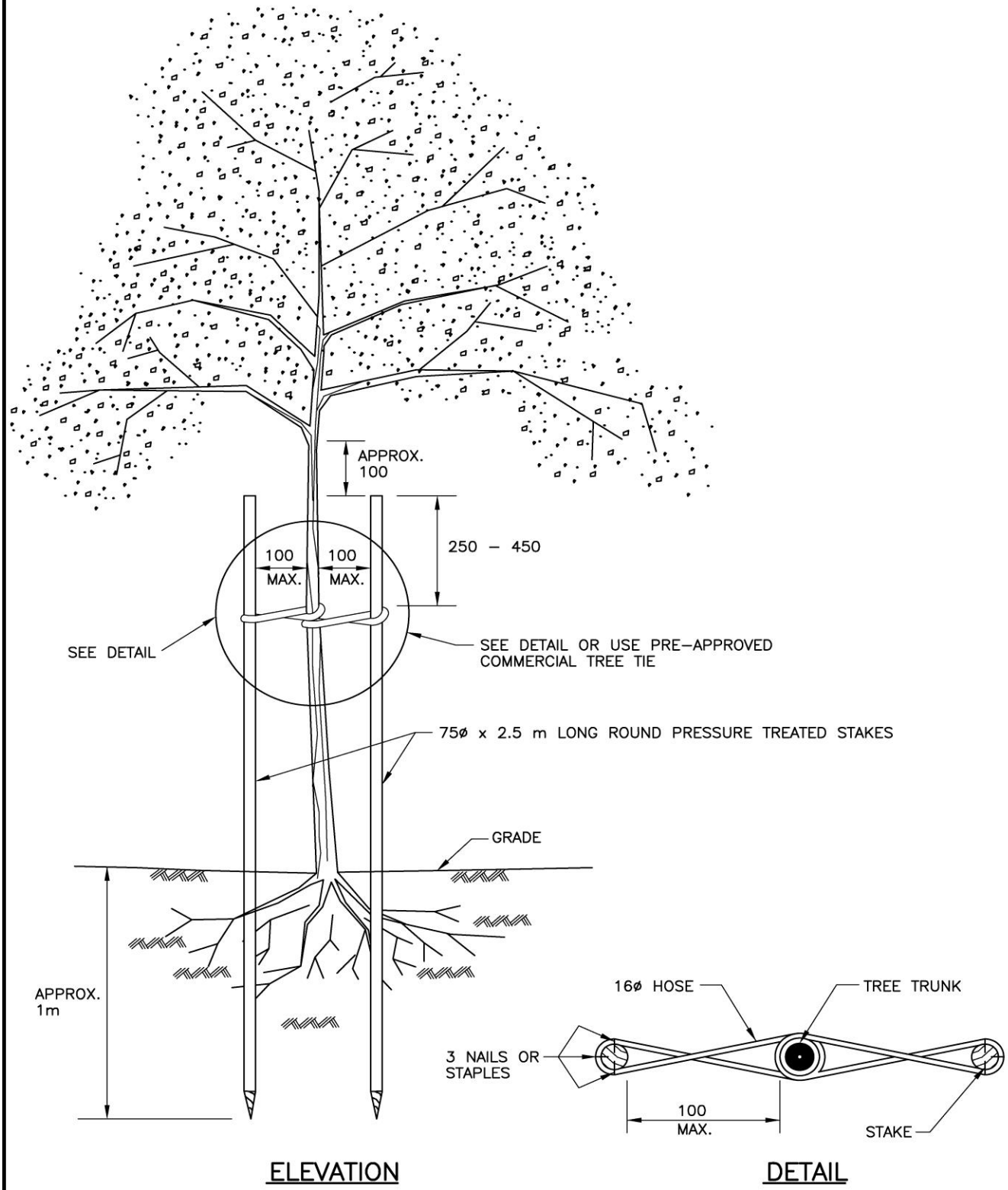


FOR CONIFEROUS TREES
WHEN SPECIFIED

NOTE:
TIGHTEN WIRE BY ADJUSTING TURNBUCKLES

NOT TO SCALE ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

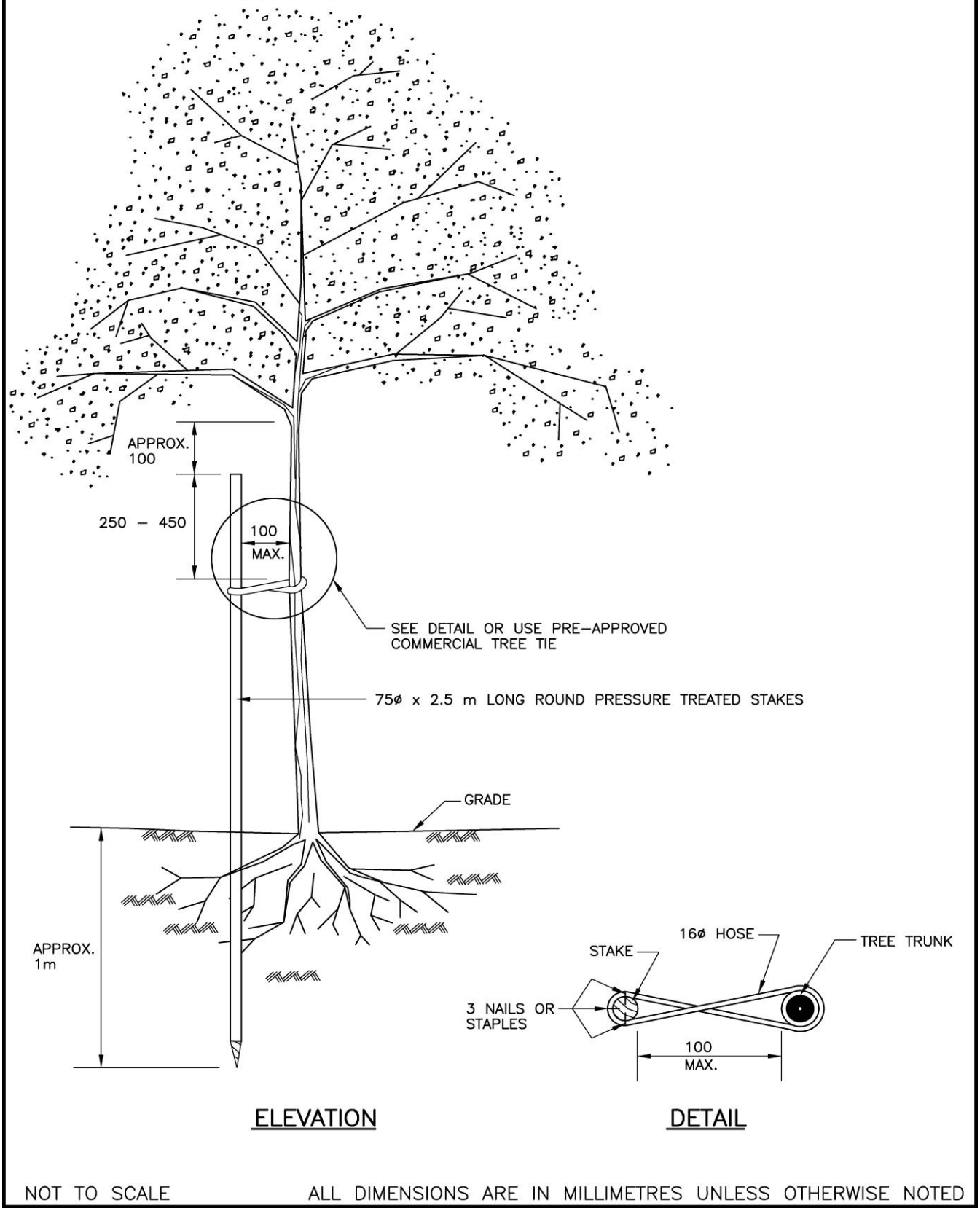
DOUBLE TREE STAKES FOR TREES OVER 3 m HIGH	SP754–05



NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SINGLE TREE STAKE FOR TREES 1.5 m TO 3 m HIGH	SP754–06

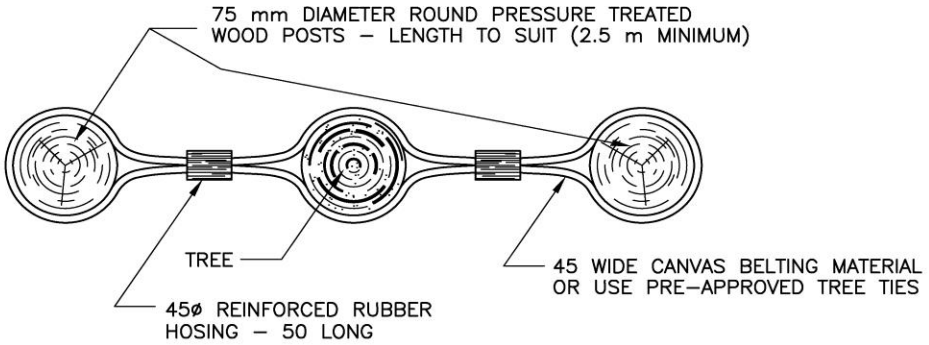


NOT TO SCALE

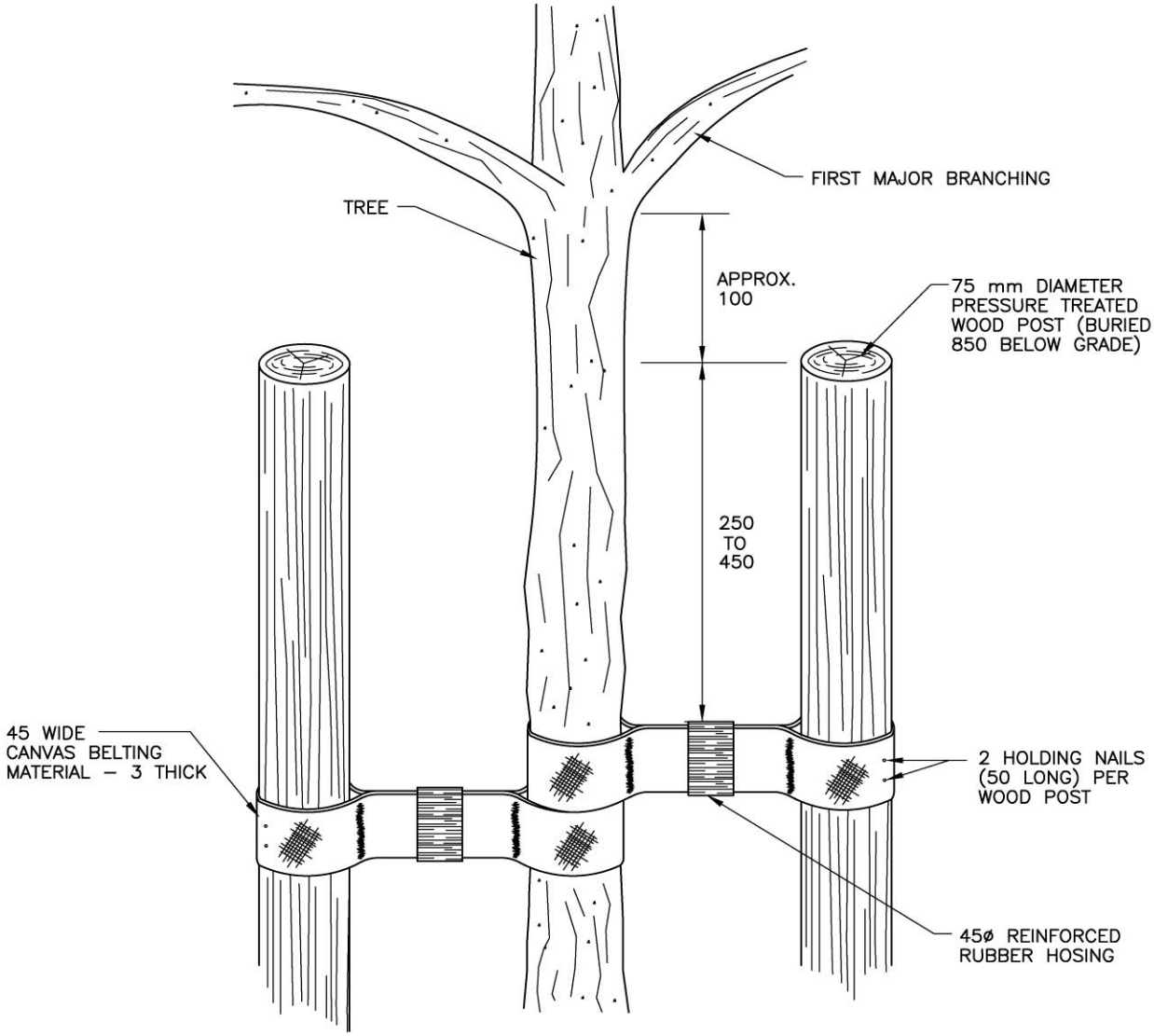
ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

DOUBLE TREE STAKES FOR BOULEVARD PLANTING AREAS

SP754–07



CROSS SECTION



NOT TO SCALE

ALL DIMENSIONS ARE IN MILLIMETRES UNLESS OTHERWISE NOTED

SECTION 757
REVEGETATION SEEDING

DESCRIPTION

757.01 Scope – This Section refers to those portions of the work that are unique to the supply and application of seed, fertilizer, mulch, tackifier, and other materials used for revegetating disturbed areas, and that are not designated for treatment under SS 754 Planting of Trees, Shrubs, and Ground Covers or SS 751 Topsoil and Landscape Grading. This Section must be referenced and interpreted simultaneously with all other Sections pertinent to the works described herein.

757.02 References

- BC Invasive Species
- BC Hydro Electrical safety resources,
- Canada Seeds Act & Seeds Regulations,
- Weed Control Act & Weed Control Regulation, and
- Canada Fertilizers Act & Fertilizers Regulations
- The Ministry's Recognized Products List

MATERIALS

757.11 Handling and Storage – All hydraulic or rolled erosion control products, seed, mulch, fertilizers and other dry materials shall be stored in a dry, weather proof storage place and shall be protected from damage by heat, moisture, weather, excessive temperatures, rodents, construction operations, or other causes until the time of seeding.

(a) All manufacturer's instructions regarding storage, handling, blending and installation shall be adhered to unless otherwise specified by the Contract.

(b) Supplier labels or other identification are not to be removed or defaced.

757.12 Seed

757.12.01 Supply of Seed – All seed specified shall be supplied by the Contractor and obtained from a recognized source.

757.12.02 Seed Type and Grade – All seed supplied either as individual species, or as a seed mix, shall comply with the requirements of the Canada Seed Act and Seeds Regulations, and the grade standards for that particular crop kind.

(a) Grass seed shall meet or exceed Common No. 1 grade prior to mixing with other species.

(b) Seed shall be free of propagules of plant species designated as noxious under the Weed Control Act & Weed Control Regulation, any BC Proposed Prohibited Species, any sweet clover and any other undesirable contaminants indicated in the Contract.

(c) Seed mixes used for general roadside revegetation, and for the general conditions and areas indicated, shall be as shown on Table 757-A and Table 757-B unless otherwise specified in the Special Provisions.

(d) When specified, specialized seed mixes, or forb, shrub or tree seed shall be supplied to the requirements of the Special Provisions.

(e) Legumes and other species palatable to wildlife shall not be included unless specified by the Contract.

757.12.03 Seed Analysis Report – The Contractor shall provide valid Certificates of Seed Analysis for each species and seed lot to be used in a mix at least seven (7) days prior to the desired blending of mixes and shipping of the seed from the seed supplier to the Contractor, for the approval by the Ministry Representative. The Certificates of Seed Analysis shall set out details of the seed as specified in the Canada Seeds Act and shall be completely free of any noxious weeds listed on the Weed Control Regulation, any BC Invasive Species, any sweet clover and any other undesirable contaminants indicated in the Contract.

The Ministry Representative will review the Certificates of Seed Analysis and, if the specifications are met, give the Contractor approval to proceed with blending and seed use.

757.12.04 Packaging and Labelling – Seed shall be supplied in the original sealed packages, with legible labels securely attached, and providing the following information:

- Supplier's name and address
- Analysis of seed mixture – the grade, and the name and percentage by weight of individual seed species
- Percentage of Pure Live Seed (PLS) for each species
- Lot number and crop year for each species in the mix
- Net weight (mass)
- Date and location of packaging

757.13 Fertilizer – Fertilizer shall comply with the provisions of the Canada Fertilizer Act and Seeds Regulations. Fertilizer shall be supplied as noted on Table 757-B and applied at the rate specified in Table 757-A unless otherwise specified in the Special Provisions. All fertilizer shall be a coated, slow-release Nitrogen type formulation.

757.14 Hydraulic Erosion Control Products (HECPs)

(a) HECPs shall either be a 100% wood fibre type or be listed on the Ministry's Recognized Products List.

(b) HECPs shall be manufactured, degradable, pre-packaged fibrous materials specifically designed for hydraulic application.

(c) HECPs shall be supplied in weather resistant packages bearing the manufacturer's label, clearly indicating the weight and product name, and shall be protected following the manufacturer's recommendations and protected from climatic conditions, including excessive sunlight to avoid packaging deterioration

(d) The fibres shall be dyed green with a water-soluble non-toxic dye that will not stain masonry, concrete, asphalt or painted surfaces, and shall be non-toxic to plant or animal life.

757.15 Tackifier

(a) Tackifier shall be an organic guar gum or starch base product specifically designed for use with HECPs and/or seeding operations.

(b) If supplied separately with a mulch product, the tackifier shall be supplied in packages bearing the manufacturer's label, clearly indicating product name, content, and application instructions.

(c) HECPs may contain a pre-mixed tackifier, which shall adhere to mulch to prevent separation during shipment and to avoid chemical agglomeration during mixing in hydraulic mulching equipment.

(d) Mulch and tackifier combination products must be premixed and bagged by the manufacturer to ensure quality control.

757.16 Water – Water used for hydraulic seeding operations shall be free of impurities that would inhibit germination and growth or may be harmful to the environment.

(a) Unless otherwise noted in the Special Provisions, the Contractor shall be responsible for securing a water source for hydraulic application of materials, including obtaining use permits under the Water Act if water is to be drawn from waterbodies, and for all cost to supply.

(b) Obtaining water from waterbodies without use permits is prohibited.

757.17 Other Materials – Where steep slopes, poor soils or sensitive areas exist, specialized HECPs including Bonded Fiber Matrix (BFM) products, Soil Amendments, including Biotic Soil Media (BSM) and other materials including Rolled Erosion Control Products (RECPs) may be specified in the Contract.

(a) Unless otherwise stated in the Contract or approved by the Ministry Representative, installation shall be in strict compliance with the manufacturer's rates and/or instructions for the slope and site.

(b) Bonded Fiber Matrix coverings, erosion control blankets, soil amendments and other materials shall be supplied to the specifications in the Special Provisions.

757.17.02 Bonded Fiber Matrix (BFM) Products – Where specified in the Contract, all hydraulically applied BFM products used shall be designed to bond with the soil surface and create a continuous, porous, absorbent and flexible erosion resistant mat to all disturbed areas. The BFM product shall be designed to stand up to multiple rainfall events for reliable erosion protection for a minimum of 6 months.

(a) The BFM product shall be composed of sterilized wood fibers and may contain organic guar gum or starch base tackifiers in addition to the proprietary components or be listed on the Ministry's Recognized Products List.

(b) All components of the BFM product shall be 100% biodegradable, non-toxic to plant or animal life and shall not form a water-impermeable crust or otherwise inhibit plant growth.

(c) The BFM product shall be designed to stand up to multiple rainfall events for reliable erosion protection for a minimum of six (6) months.

757.17.03 Biotic Soil Media (BSM) – Where specified for use, the BSM shall be designed as an alternative to topsoil and/or compost to accelerate development of depleted soils/substrates with low organic matter, low nutrient levels and limited biological activity.

(a) BSM products used must be non-toxic with fibers that have been phytosanitized, prior to the introduction of soil building components, to eliminate potential weed seeds and pathogens.

(b) BSM products shall not be used for erosion control and shall be applied to enhance vegetation establishment and growth only.

(c) Where erosion control is required, the BSM shall be hydraulically installed in a two-phase process with the BSM installed first with the seed, then covered with the mulch and tackifier or BFM product. BSM products may also be applied by hand in conjunction with broadcast seeding.

EQUIPMENT

757.21 General – Equipment used shall be capable of applying the materials listed in the Special Provisions uniformly over the designated areas.

Equipment shall not cause soil rutting or other site damage.

757.22 Hydraulic Seeding/Mulching Equipment – Equipment shall have the tank volume identified by an

identification plate or sticker, which shall be affixed in plain view.

The hydraulic machine shall be capable of sufficient agitation to mix the materials into a homogenous slurry, and to maintain the slurry in a homogeneous state until and during application.

Equipment shall be adequately sized to the task, to complete work efficiently within the time frame specified, and to permit application of materials without excess water being applied, or undue time lapse between operations. Hydraulic machines should be capable of producing slurry viscosities containing approximately 18 to 30 kg of mulch per 500 litres of water.

Extension hoses or pipes shall be provided to reach areas not accessible from the hydraulic seeder.

CONSTRUCTION

757.31 Scheduling – Work shall be scheduled to ensure a minimum duration of on-Site storage of materials, minimum compaction of topsoil, and prompt seeding/mulching operations.

The work shall be coordinated with the schedule of other trades, and be well integrated with specific requirements such as Sediment and Drainage Management Plans and best practices, which may be provided for any given project.

757.32 Protection – Existing Site equipment, roadways, landscaping, reference points, monuments, markers, utilities, and structures shall be protected from damage by hydraulic application method. No overspray is to occur into waterbodies or environmentally sensitive areas. When necessary to ensure protection of these areas, dry, hand broadcasting of materials will be employed.

Areas treated with HECPs shall be protected from foot and vehicle traffic, grazing and other disturbances. Any damaged areas that result in disturbance of the continuous mat of HECP materials and exposed soil shall be repaired within 24 hours of notification of the damage.

757.33 Substrate and Seedbed Preparation – Prior to installation of HECPs, the Contractor will all ensure all slopes are geotechnically stable and have been designed to divert runoff away from the face of the slope where required.

(a) All pre-existing rills or gullies shall be repaired prior to installation, and slopes shall be suitably roughened to the Ministry Representative's satisfaction.

(b) Slopes shall not be left in a smooth or glazed condition that functions to prevent water and plant root infiltration.

(c) Slope roughening techniques may include, but not be limited to, track walking, imprinting, harrowing, or placement of woody debris.

(d) Indentations must be horizontally installed to the slope rather than vertically, to ensure they function to trap water and seed rather than accelerate erosion.

(e) Any slope interruption devices specified shall be correctly installed prior to application of the HECP.

757.34 Timing of Material Application – Material application shall be carried out in accordance with the construction schedule and shall occur as soon as possible following the disturbance.

(a) Reseeding during extreme drought or heat may result in failure and should be avoided.

(b) Hydraulically applied erosion control products shall not be applied to saturated soils or substrates, and shall not be installed immediately before, during, or after rainfall, such that the slurry will have the opportunity to dry for up to 24 hours after installation, unless otherwise approved by the Ministry Representative or permitted by the Manufacturer's specifications for the product being used.

757.35 Methods – The methods chosen for material application shall be at the Contractor's discretion, unless otherwise specified in the Special Provisions.

757.36 Rates of Application – Unless otherwise specified by the manufacturer, standard material application rates for mulch and tackifier shown in Table 757-A shall be followed. Fertilizer shall be applied as recommended by soil testing or following Table 757-B.

(a) Site conditions including allowances for increased surface area due to roughening or scarification during substrate and seedbed preparation shall be taken into account when determining quantities of materials required to reach the specified application rates.

(b) Application rate may need to be increased by up to 20% for tracked surfaces.

757.37 Record of Application – The Contractor shall maintain a record of all pertinent application information on the form accepted by the Ministry Representative. Refer to Figure 757-1.

757.38 Application Method for Mechanical Drop or Broadcast Dry Seeding – Seed shall be applied in two intersecting directions and overlap adjoining ground cover by 300 mm, except where conditions dictate seeding in one direction only.

Dry seeding must occur during spring or fall timing windows and shall be installed at a rate of 30 kg/ha. Seed shall be raked or rolled into the soil after sowing, and where necessary, may be covered with a layer of thin and continuous layer of straw to help protect the seed from wildlife and prevent erosion

757.39 Hydraulic Application of Materials

757.39.01 General – The hydraulic machine shall be operated in compliance with Ministry safety standards including those detailed in the BC Hydro Electrical safety resources and the Ministry`s Traffic Management Manual for Work on Roadways.

(a) Materials shall not be sprayed on objects not expected to support plant growth.

(b) Existing Site equipment, roadways, landscaping, reference points, monuments, markers, utilities, and structures shall be protected from damage by hydraulic application method.

(c) No overspray is to occur into waterbodies or environmentally sensitive areas.

(i) HECPs are not intended to be applied in channels, swales, or other areas where concentrated flows are anticipated unless installed in conjunction with RECPs. When necessary to introduce vegetation to these areas, dry, hand broadcasting of materials shall be employed.

(d) The Contractor shall be responsible for any overspray or damages incurred during hydroseeding. Any overspray or damage shall be made good at no cost, to the satisfaction of the Ministry Representative.

757.39.02 Mixing

(a) The required quantities of seed, fertilizer, mulch, tackifier and other material shall be charged into the tank accurately by weight or by an acceptable system of mass calibrated volume measurement, following the manufacturer's specifications.

(b) The materials shall be thoroughly mixed into homogeneous water slurry for a minimum of 10 minutes prior to application for each load.

(c) All seed shall be added last when mixing.

(i) Pellet inoculated seed shall be applied immediately after placement into tank, and if this is not possible, dry application methods must be used.

(ii) Other seed shall not be left in the tank for unreasonable lengths of time prior to application, i.e. – exceeding one or two hours, particularly when in contact with fertilizer solution.

(d) The Ministry Representative will determine if Seed that remains in the tank for periods longer than specified can be used. Rejected seed shall be replenished with fresh stock.

757.39.03 Application – HECPs shall be applied to form an even, uniform mat blended 150 mm into adjacent vegetated areas or previous mulch applications. HECPs shall be applied from opposing directions, except where conditions dictate seeding in one direction only, to achieve soil coverage.

757.40 Related Work – Additional related work such as the application of Rolled Erosion Control Products (RECPs), compost blankets, or other coverings, and harrowing or discing of soil following material application, shall be as specified in the Special Provisions and be installed in accordance with the manufacturer's specifications.

757.41 Clean-up – All surplus and waste materials resulting from seeding operations shall be removed from the job site after empty product containers have been inspected by the Ministry Representative.

Hydraulic seeding and/or mulching overspray that may cause problems on areas or objects not designated for re-vegetation shall be removed in an appropriate manner.

757.42 Conditions for Acceptance – Treated areas will be accepted by the Ministry when the following conditions have been met.

(a) Application rates meet or exceed the specified rates.

(b) Treated areas are not thin with bare patches, or uneven in distribution.

(c) Empty containers of materials used during the work are stored neatly on Site for inspection by the Ministry Representative.

757.43 Guarantee & Repairs – The Contractor shall restore disturbed areas beyond the cut slopes at the direction of the Ministry Representative.

Seeded areas that show thin application or bare spots shall be re-treated with the specified materials at the earliest opportunity, weather and season permitting.

No additional payment will be made for the repairs.

MEASUREMENT

757.81 General – Revegetation Seeding shall be measured by the area treated, to the nearest tenth of a hectare (0.1 ha). The treated areas will be calculated by actual measurement along the slope, and within the right-of-way, or as directed by the Ministry Representative.

PAYMENT

757.91 General – Payment for REVEGETATION SEEDING will be at the Unit Price per hectare bid for "Revegetation Seeding" in Schedule 7, as measured in place. The Contract Unit Price shall be accepted as full compensation for the work described and all work subsidiary and incidental thereto for which separate payment is not elsewhere provided.

Figure 757-1: Daily Seeding / Application Record

SHEET #_____

PROJECT #_____

DAILY SEEDING/APPLICATION RECORD

DATE	LOAD #	AREA COVERED IN HECTARES				SEED APPLIED* (IN KG.)			OTHER MATERIALS* (IN KG.)				REMARKS
		GRASS	MULCH	REFERT		GRASS	RYE	OTHER	FERT	MULCH	TACK	MISC.	
	1												
	2												
	3												
	4												
	5												
	6												
	7												
	8												
	9												
	10												
	11												
	12												
	13												
	14												
	15												
	16												
	17												
	18												
	19												
	20												
Total of Sheet													

Provide details of seed mixes and other materials, as required, in this space:

CERTIFIED CORRECT:

_____ Date: _____
Contractor Representative (or delegate)

Received by the Ministy:

_____ Date: _____
Ministry Representative (or delegate)

APPENDIX 757-A

Standard Grass Seed Mixes and Application Rates

757-A.01 Seed Mix Application Rates – Unless otherwise specified by the manufacturer, standard application rates are as specified in Table 757-A.

Table 757-A: Application Rates for Standard Seed Mixes

Product	Standard Application Rate (kg/ha)
Seed Mix (Table 757-B)	75
Nurse Crop Grass (when specified) • Fall Rye • Axcella Annual Rye	 50 25
Fertilizer	300
Wood Fibre Mulch	1500
Tackifier	Per Manufacturer's Instructions

757-A.02 Landscaped or Turfgrass Areas – Where seeding for landscaped or turfgrass areas is required, the mixture shall be of turf-type: Perennial ryegrass, Kentucky bluegrass, and Red Fescue, unless otherwise specified. Turfgrass mix shall be composed of any two or more of the above species mixed in combinations such that no single kind constitutes more than 75% of the grass in the turf.

757-A.03 Fertilizer – Fertilizer N-P-K proportions shall be as recommended by soil testing (preferred) or by following Table 757-A.

757-A.04 Percipitation Values – Average annual precipitation values may be found on-line, including from Environment Canada at:

- http://climate.weather.gc.ca/climate_normals/index_e.html
- http://climate.weather.gc.ca/historical_data/search_historic_data_e.html

Table 757-B: Standard Grass Seed Mixes for Revegetation of BC Roadsides

Climatic Area	Standard Mixes (by weight)	Application
South Coast	**Vancouver Island / Coast Mix** Canada bluegrass 5% Creeping red fescue 29% Hard fescue/Sheep fescue 17% Perennial ryegrass 37% Redtop 3% Timothy 9%	General seeding coastal locations where mean annual precipitation is > 90 cm. Fertilizer: 16-32-6, or pre-approved equivalent
	Interior Forestland Mix Annual ryegrass 11% Canada bluegrass 3% Hard fescue/Sheep fescue 13% Intermediate wheatgrass 50% Perennial ryegrass 22% Redtop 1%	General seeding inland where mean annual precipitation is > 50 cm. Fertilizer: 16-32-6, or pre-approved equivalent
	Interior Dryland Mix Crested Wheatgrass 40% Hard Fescue 15% Slender Wheatgrass 20% Tall Wheatgrass 25%	General seeding inland where mean annual precipitation is < 30 cm. Fertilizer: 16-32-6, or pre-approved equivalent

Climatic Area	Standard Mixes (by weight)		Application
Thompson – Okanagan	**Interior Forestland Mix**		
	Annual ryegrass	11%	General seeding inland where mean annual precipitation is > 50 cm. Fertilizer: 22-11-11, or pre-approved equivalent
	Canada bluegrass	3%	
	Hard fescue/Sheep fescue	13%	
	Intermediate wheatgrass	50%	
	Perennial ryegrass	22%	
	Redtop	1%	
	Interior Dryland Mix		
	Crested Wheatgrass	40%	General seeding inland where mean annual precipitation is < 30 cm. Fertilizer: 22-11-11, or pre-approved equivalent
	Hard Fescue	15%	
	Slender Wheatgrass	20%	
	Tall Wheatgrass	25%	
	Alkaline Tolerant Blend		
	Canada Bluegrass	15%	General seeding in alkaline soils. Fertilizer: 22-11-11, or pre-approved equivalent
	Crested Wheatgrass	35%	
	Hard Fescue	25%	
	Sherman Big Bluegrass	25%	
Kootenays	**Interior Forestland Mix**		
	Annual ryegrass	11%	General seeding inland where mean annual precipitation is > 50 cm. Fertilizer: 22-11-11, or pre-approved equivalent
	Canada bluegrass	3%	
	Hard fescue/Sheep fescue	13%	
	Intermediate wheatgrass	50%	
	Perennial ryegrass	22%	
	Redtop	1%	
	Kootenay Dryland		
	Canada bluegrass	3%	General seeding inland where mean annual precipitation is < 50 cm. Fertilizer: 22-11-11, or pre-approved equivalent
	Crested wheatgrass	24%	
	Hard fescue/Sheep fescue	13%	
	Redtop	1%	
	Tall wheatgrass	59%	
Northern **(Prince George Area)**	**North East General Mix**		
	Creeping red fescue	25%	General seeding inland where mean annual precipitation is > 50 cm. Fertilizer: 26-16-8, or pre-approved equivalent
	Perennial ryegrass	30%	
	Tall fescue	30%	
	Timothy	15%	
	North East Dryland Mix		General seeding inland where mean annual precipitation is < 50 cm. Fertilizer: 26-16-8, or pre-approved equivalent
	Creeping red fescue	22%	
	Crested wheatgrass	38%	
	Intermediate Wheatgrass	40%	